

Primary/SV62 + TDA4+A-PHY – B2

V0.1 - DRAFT
Based on B1 with A-PHY V1.41

Revised ETHSW Reset scheme to shorten Ignition to ETHSW startup time
Added MCU EEPROM Write Protect control by MCU + P.U./P.D.
Added CAN transceivers` Error pin connection to MCU

ETHSW Reset Logic OPEN-TBD

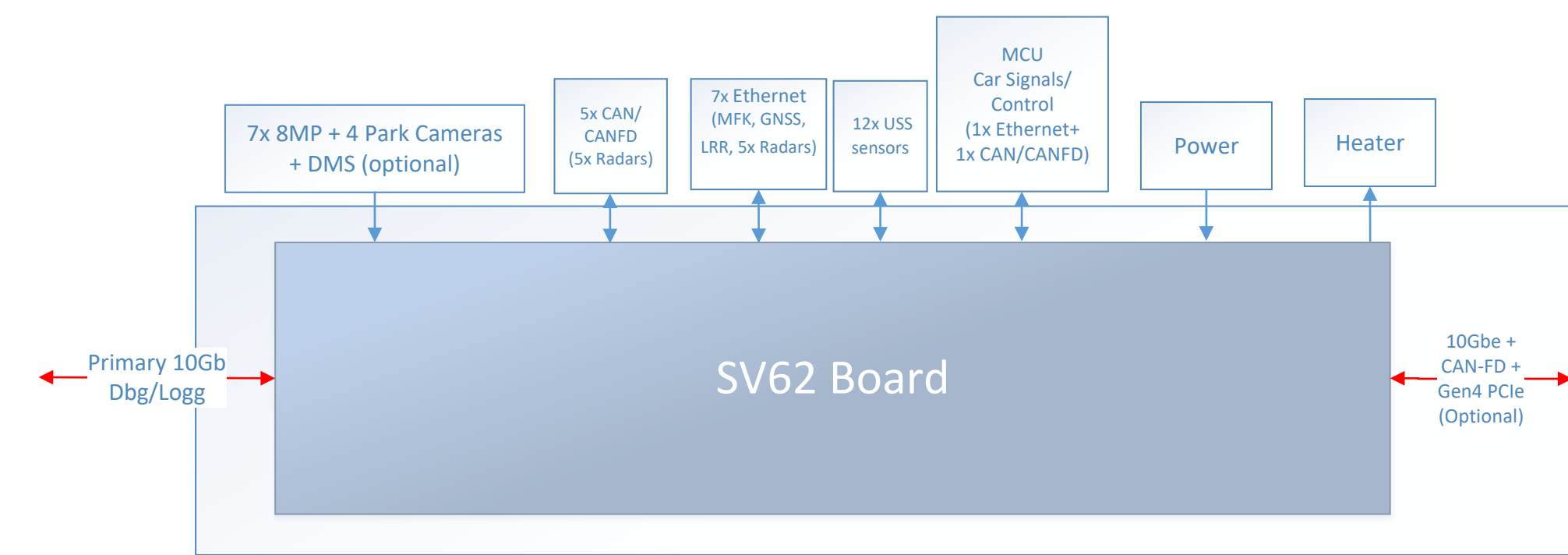
SV62 1 BOX with TDA4 - Draft suggestion

Assumptions:

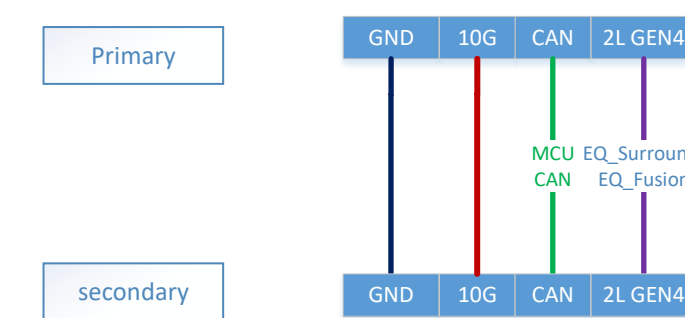
1. IP67 support
2. LRR – 100Mb ETH on shielded connector
3. Valeo PECU solution inside SV62 ECU
4. Wake over CAN
5. Support TC10
6. Support MACsec
7. TDA4 performs both PECU and video encoding functionality

Platform View

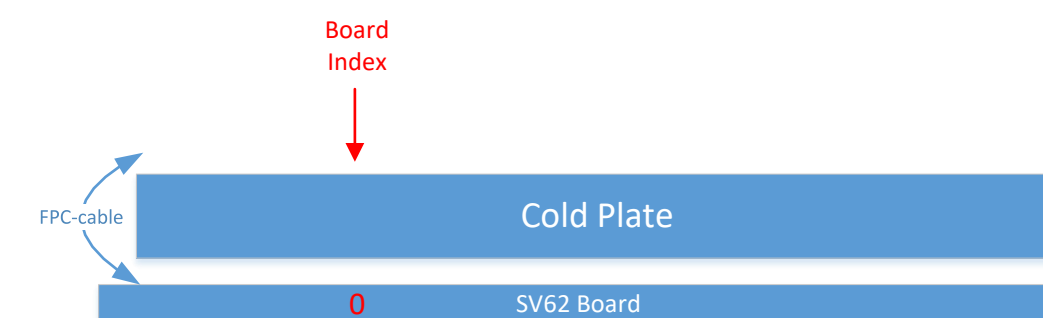
SV62 with TDA4 System Connectivity



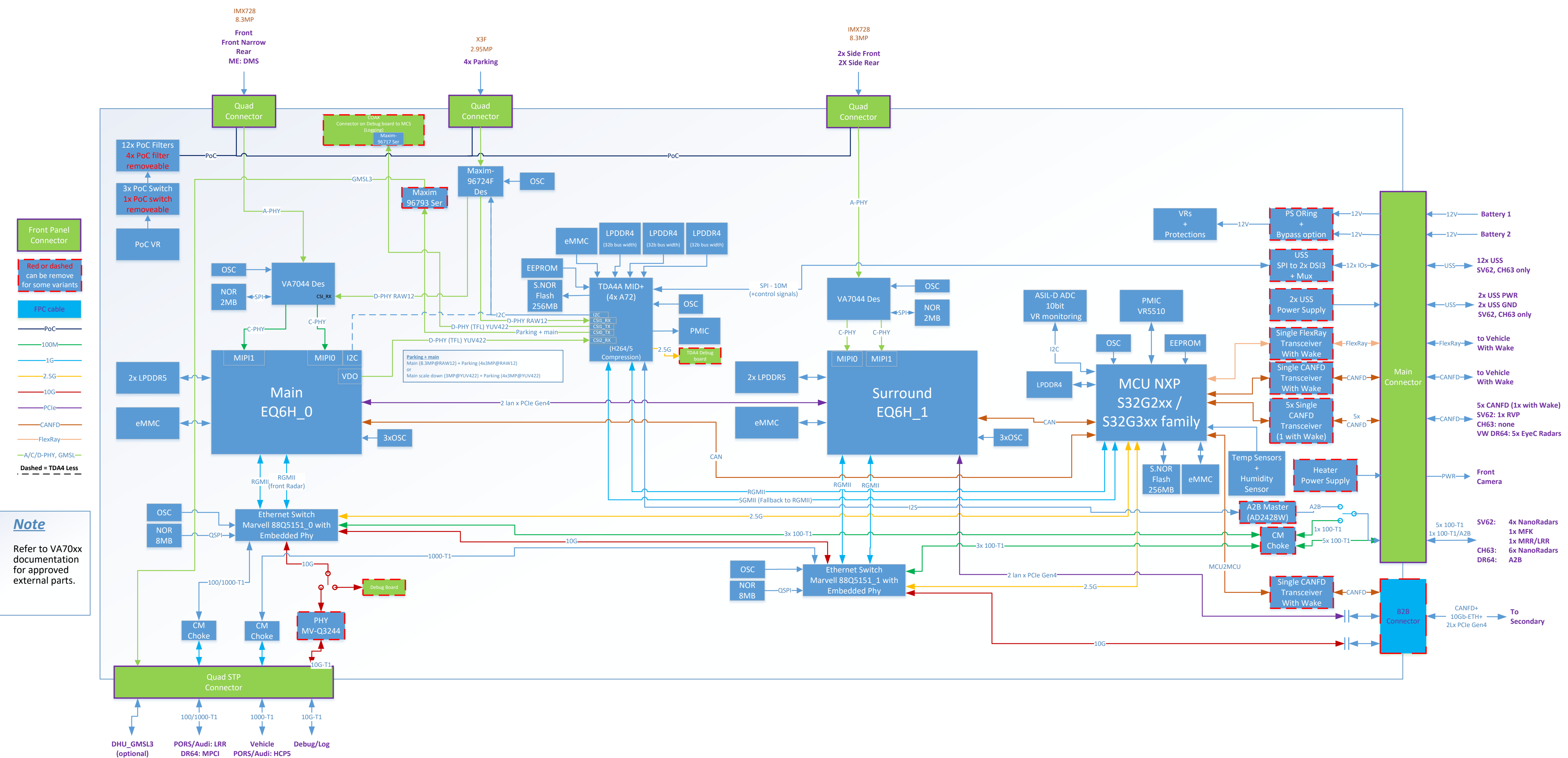
FPC-cable Connectivity

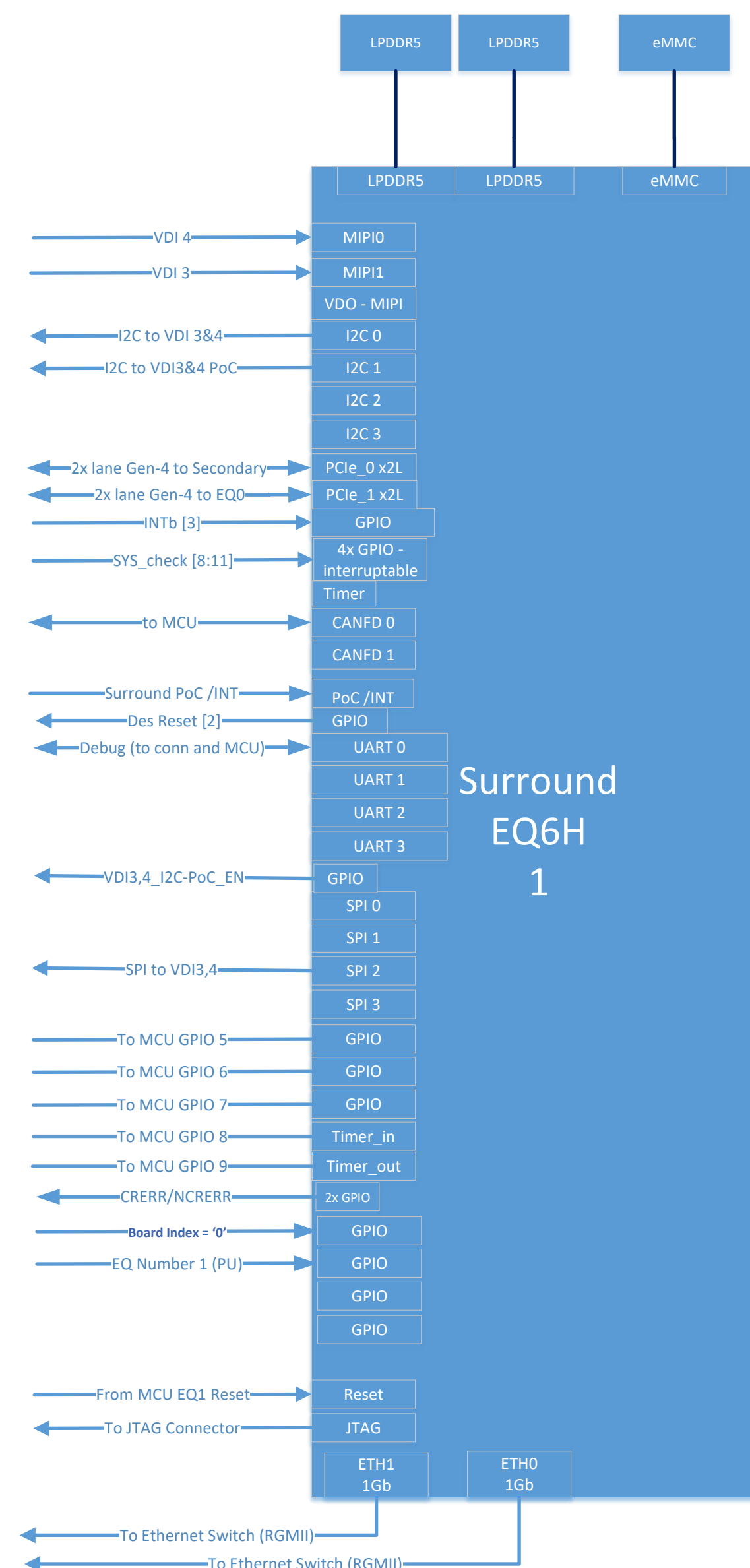
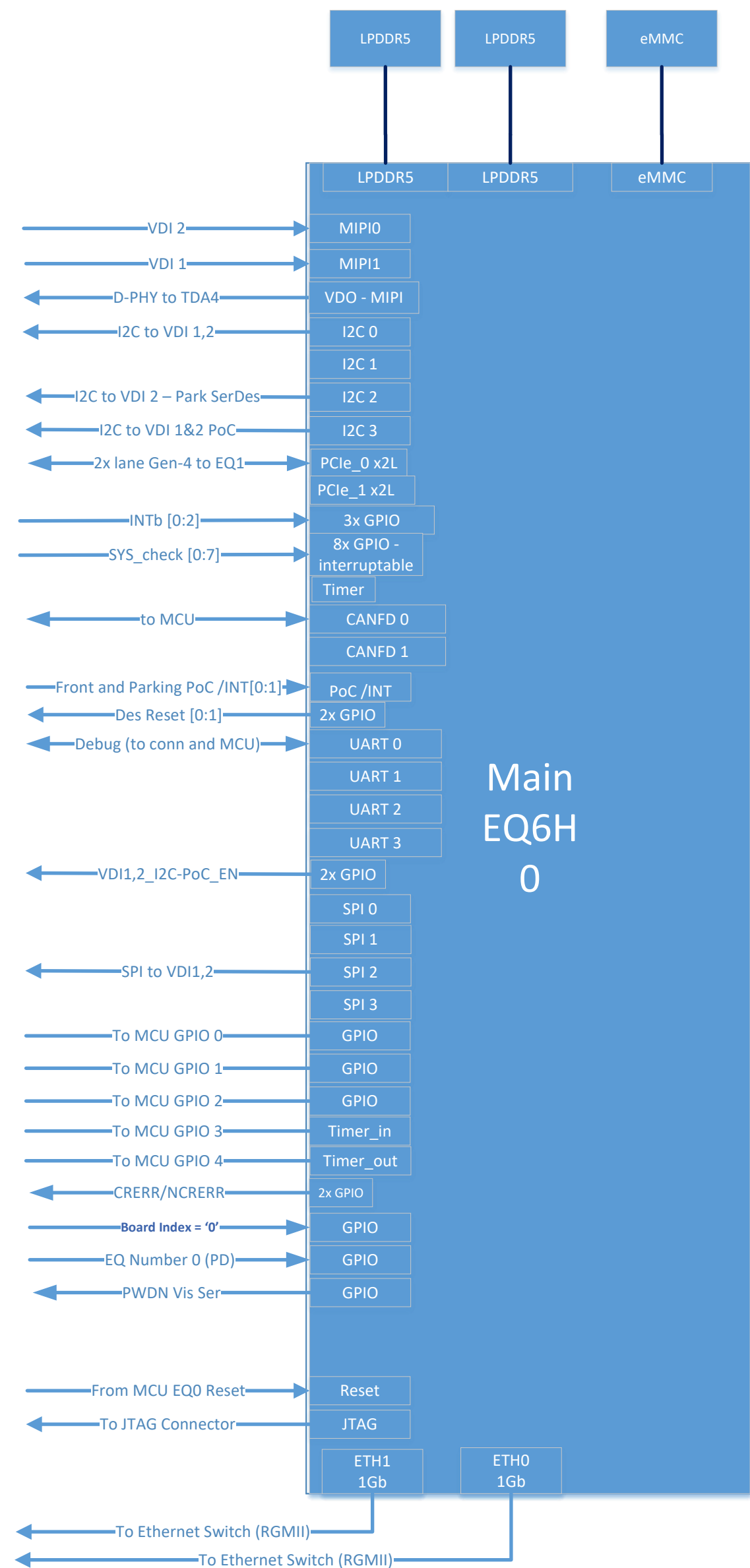


Configuration for SV62



Primary Board High-Level Block Diagram



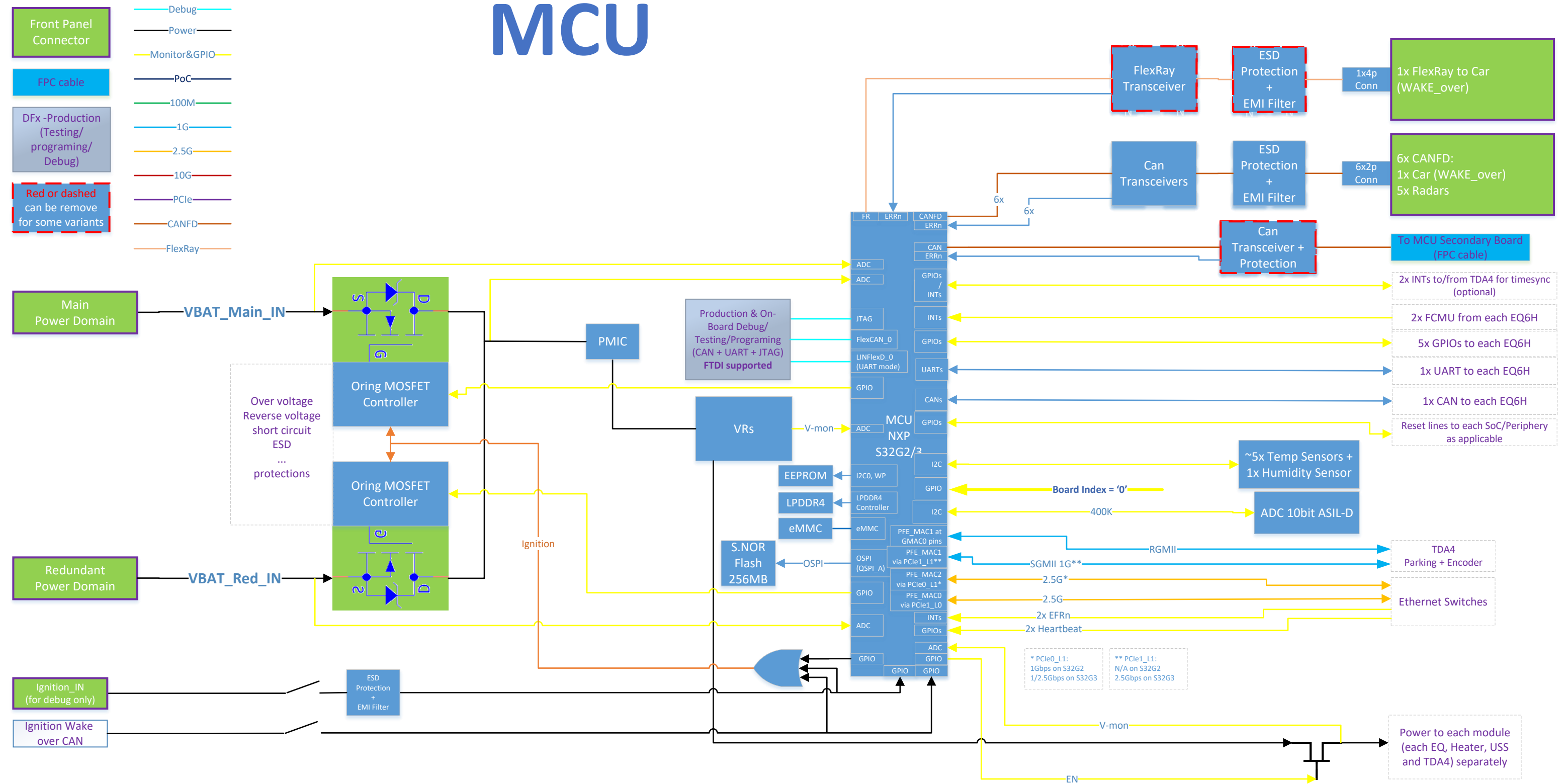


- DFx:
- JTAG chain
 - UART0
 - MCU CAN0

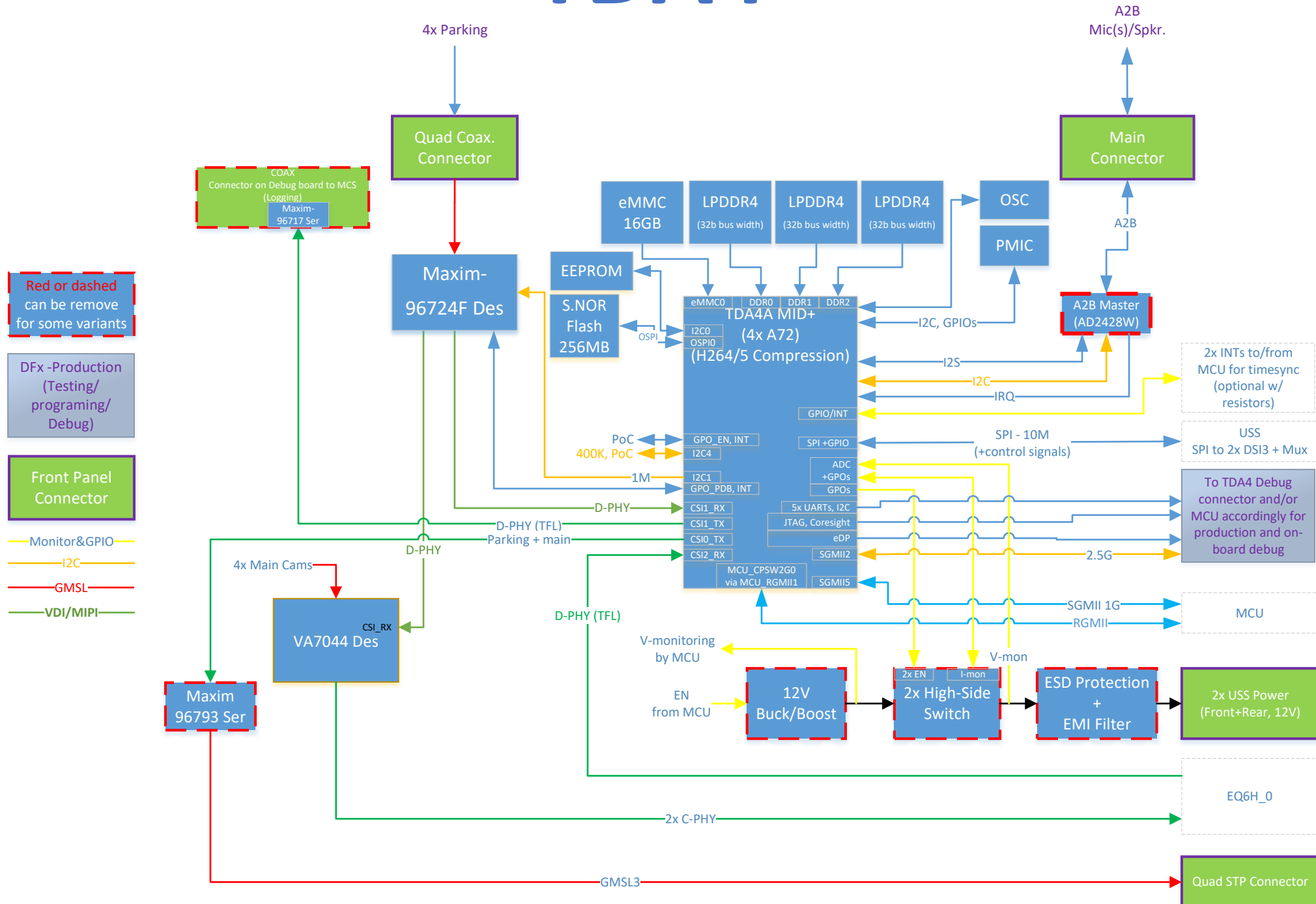
FTDI supported

Note

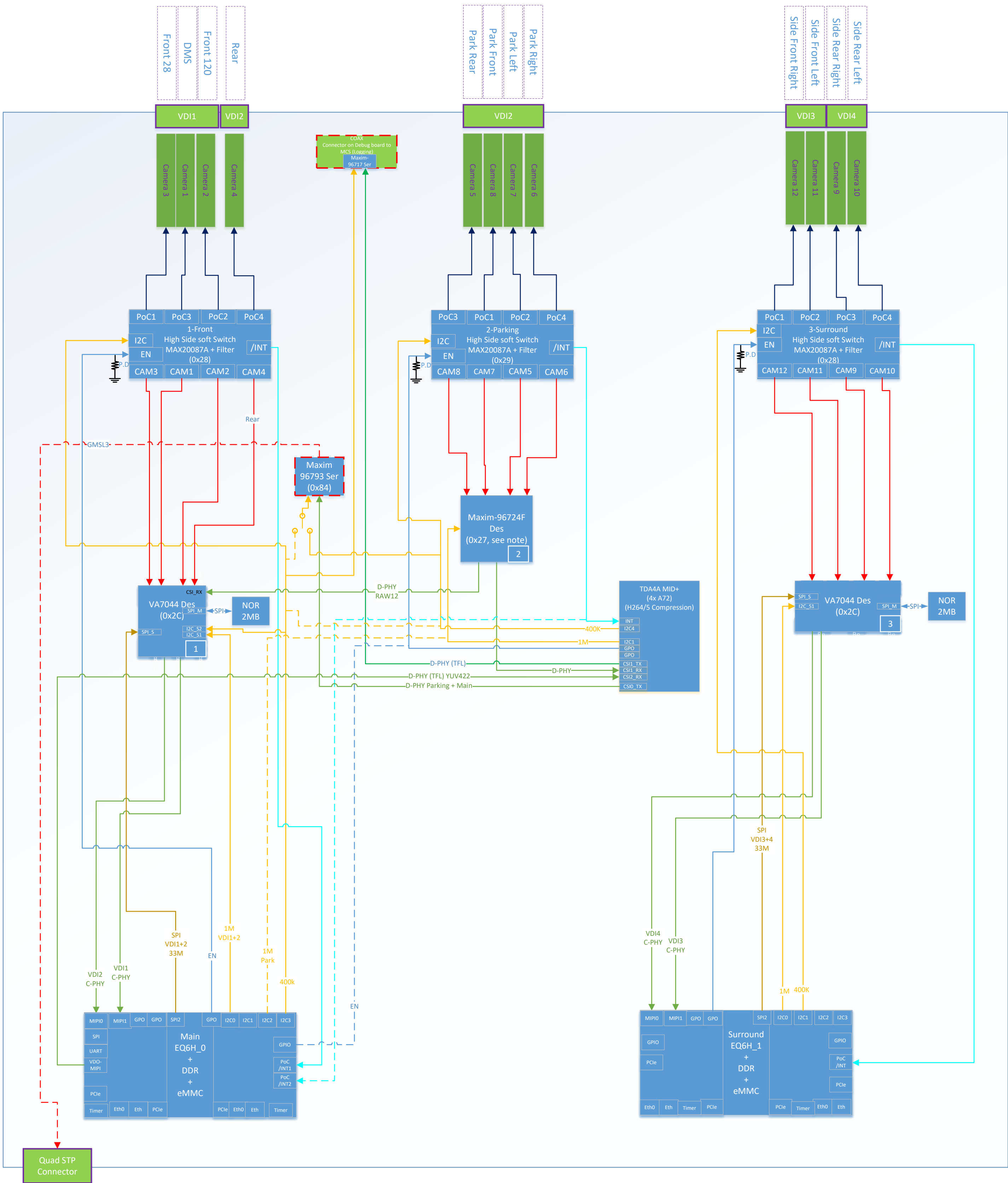
EEPROM Write Protect shall have an ICT pad, pull-up, pull-down, and MCU control options.



TDA4



I2C, SPI, PoC Master



Note

I2C addresses are presented as 7-bits (R/W bit is not appended).

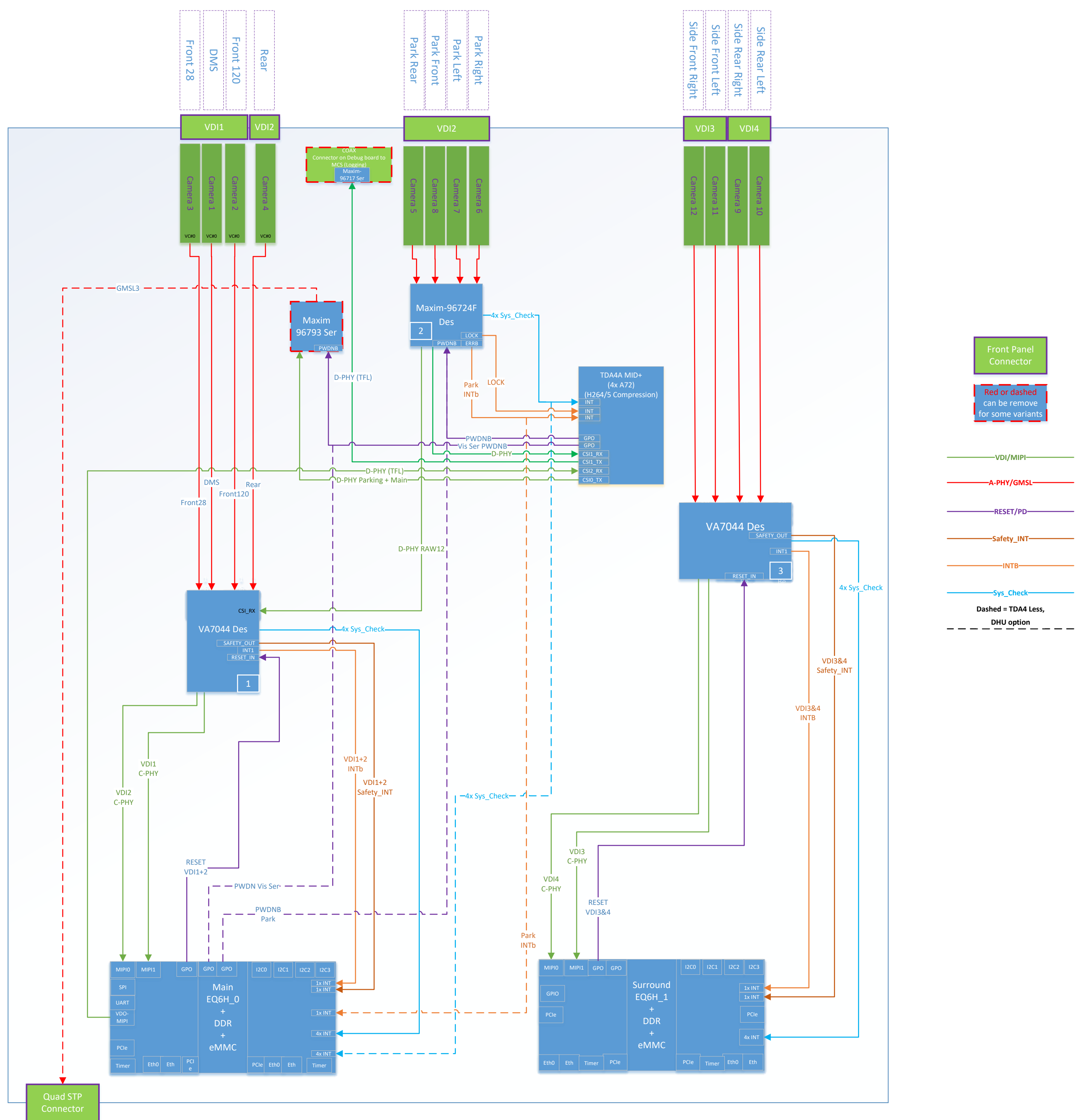
Note

Refer to VA70xx documentation for recommended PoC design.

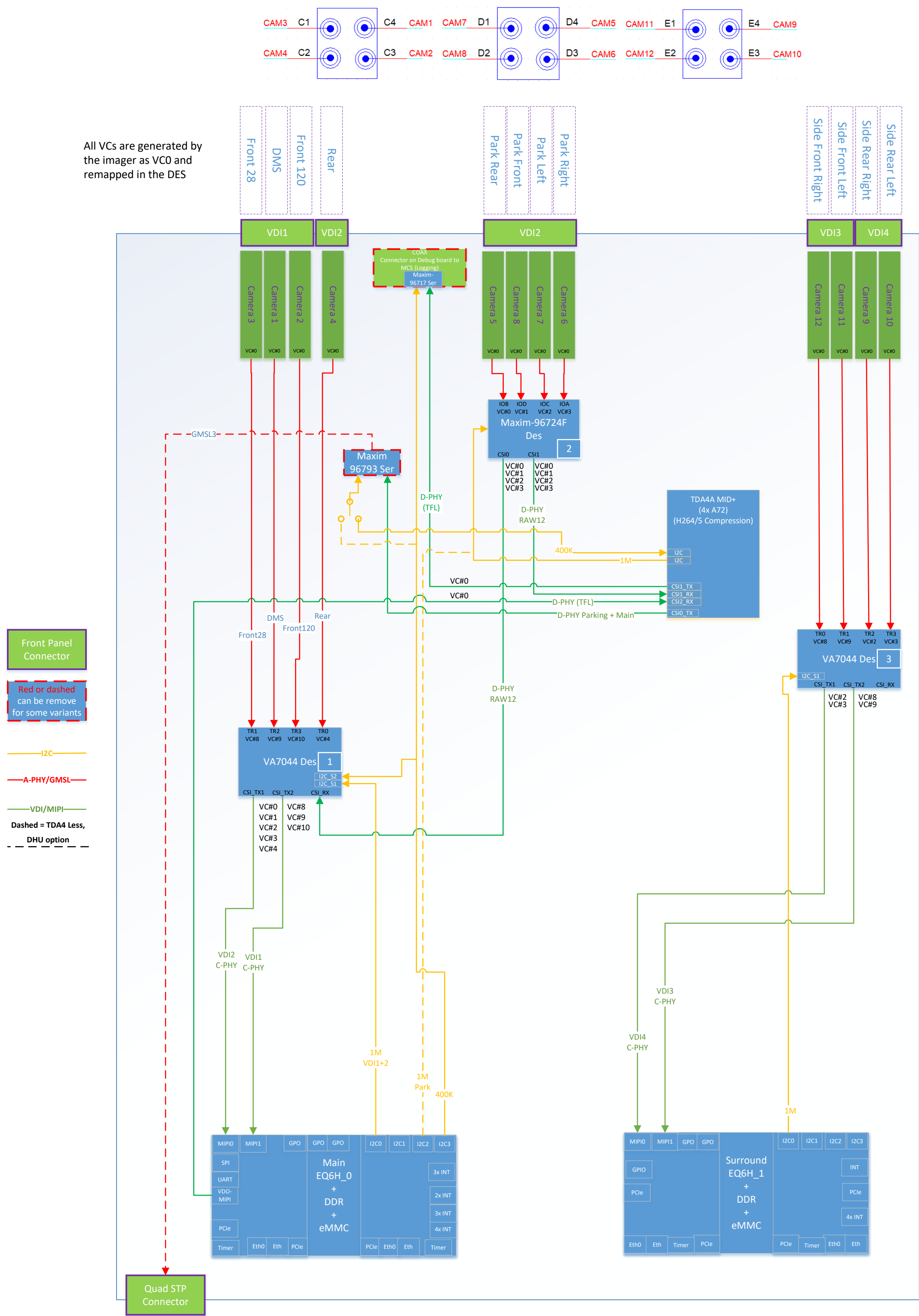
Note

VA70xx: Please add a test point to K16 and L18 for UART interface to enable Valens to debug the board.

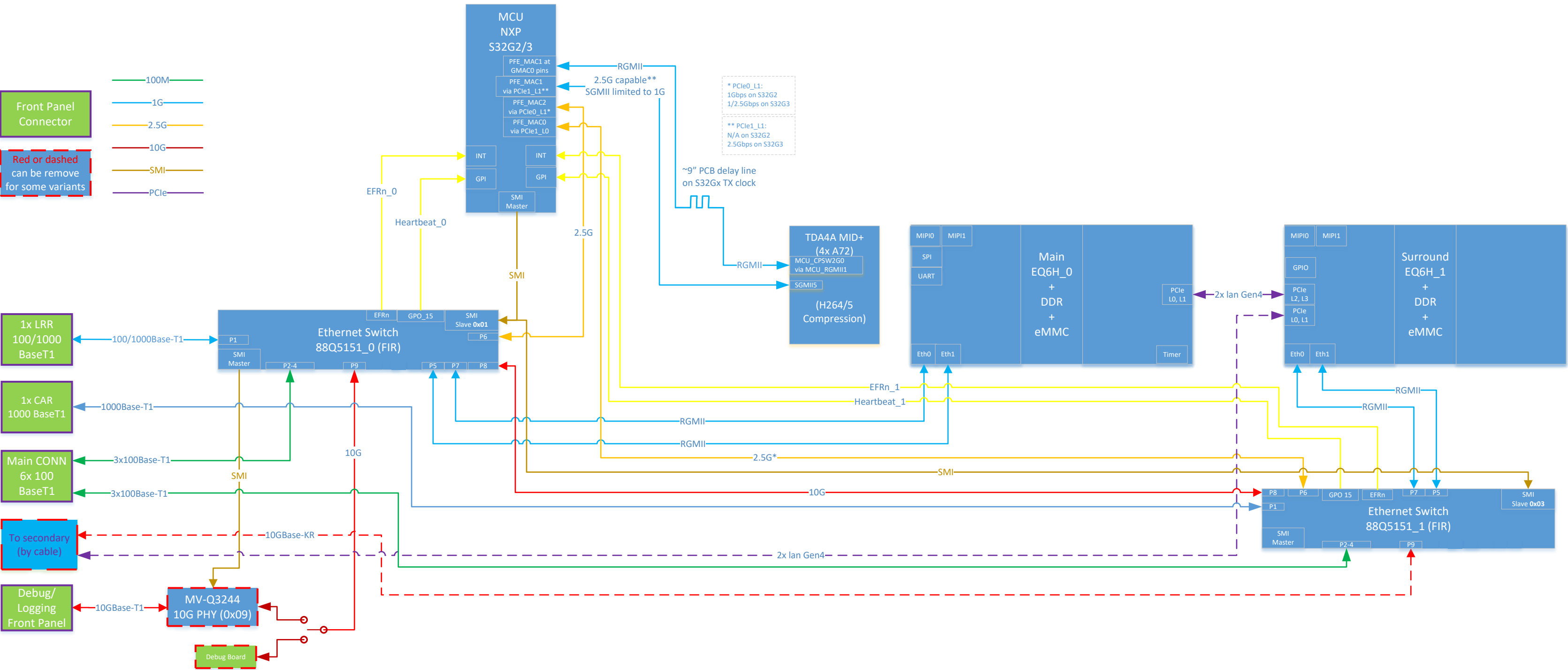
Reset/PD+INTb+Sys_Check Master



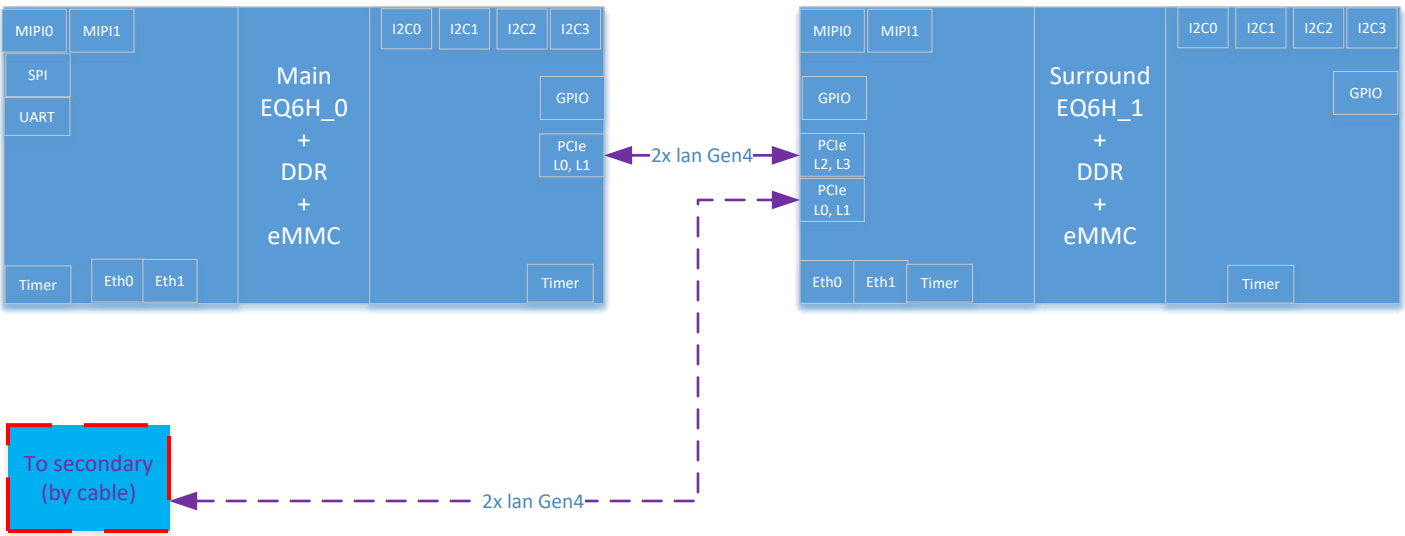
Camera RX+VC Topology



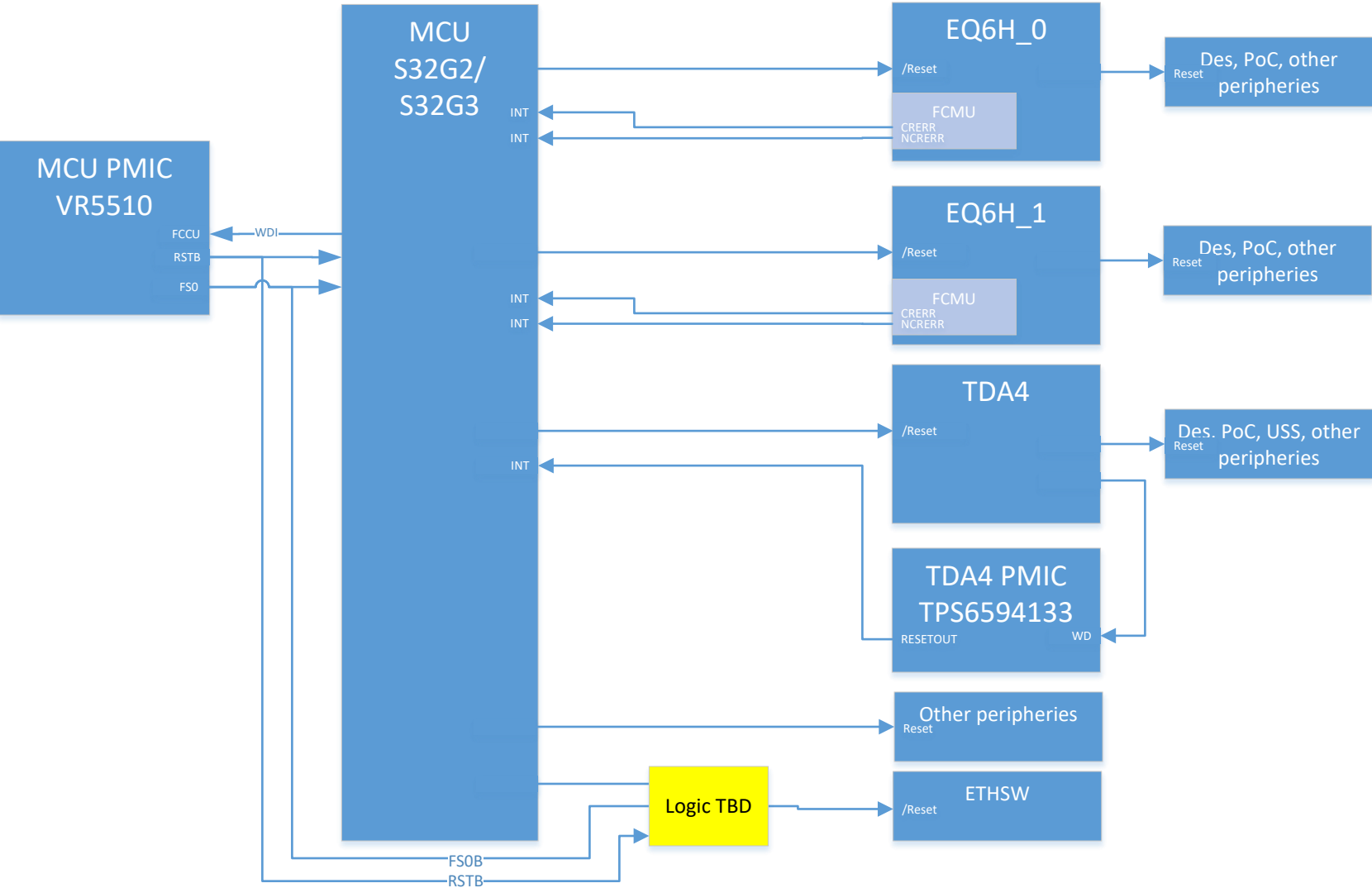
ETH+PCIe Master



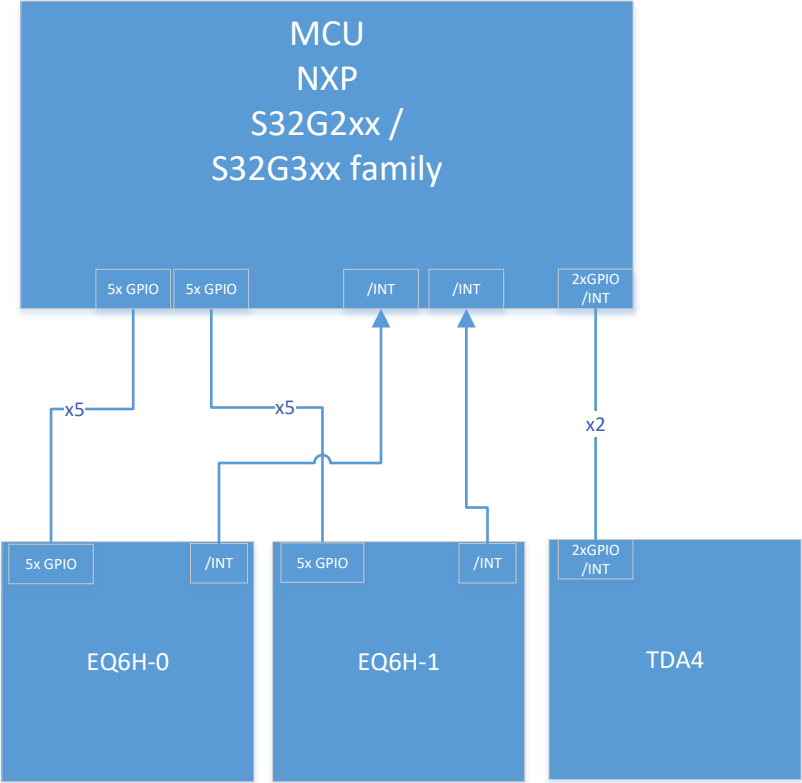
PCIe



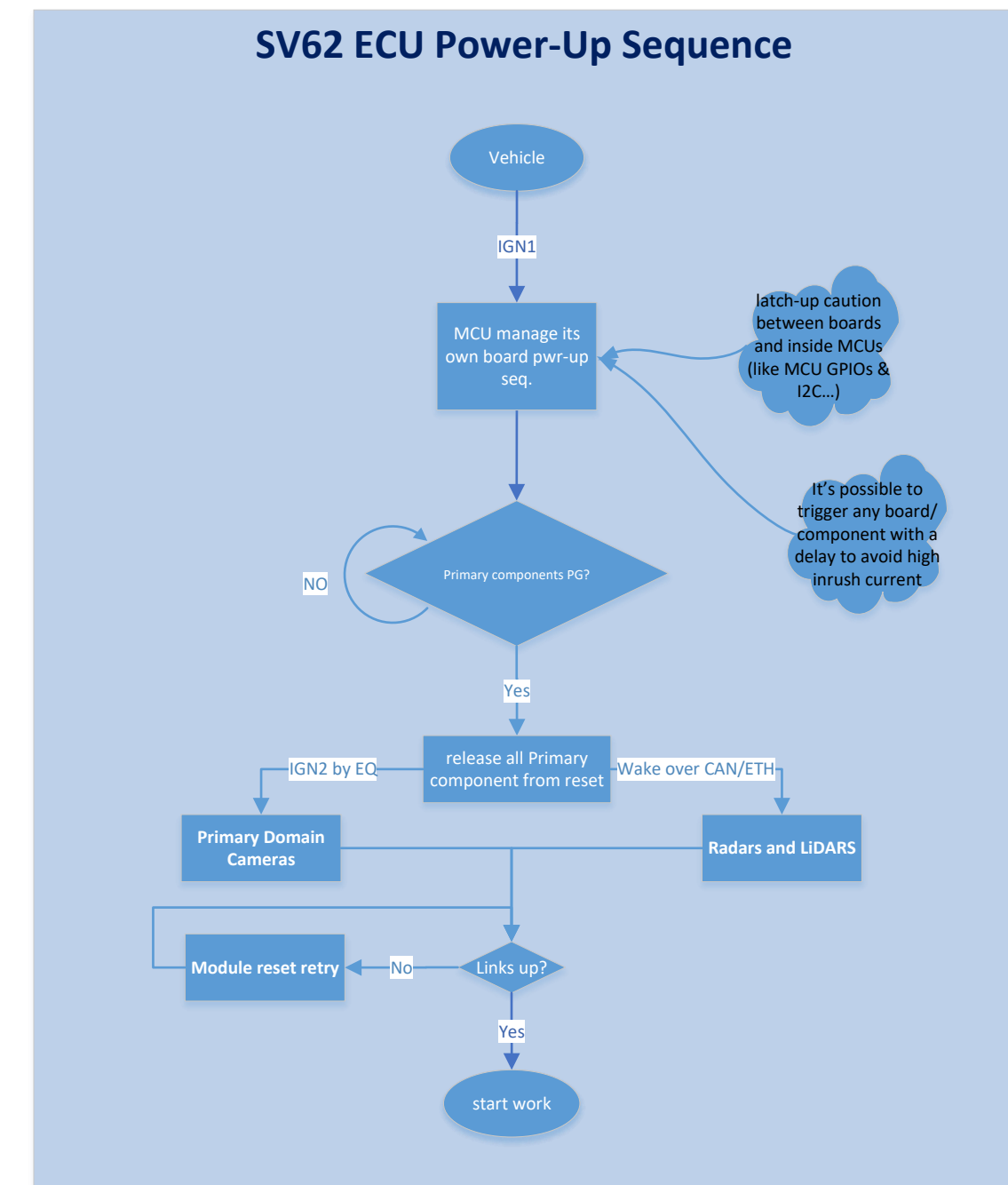
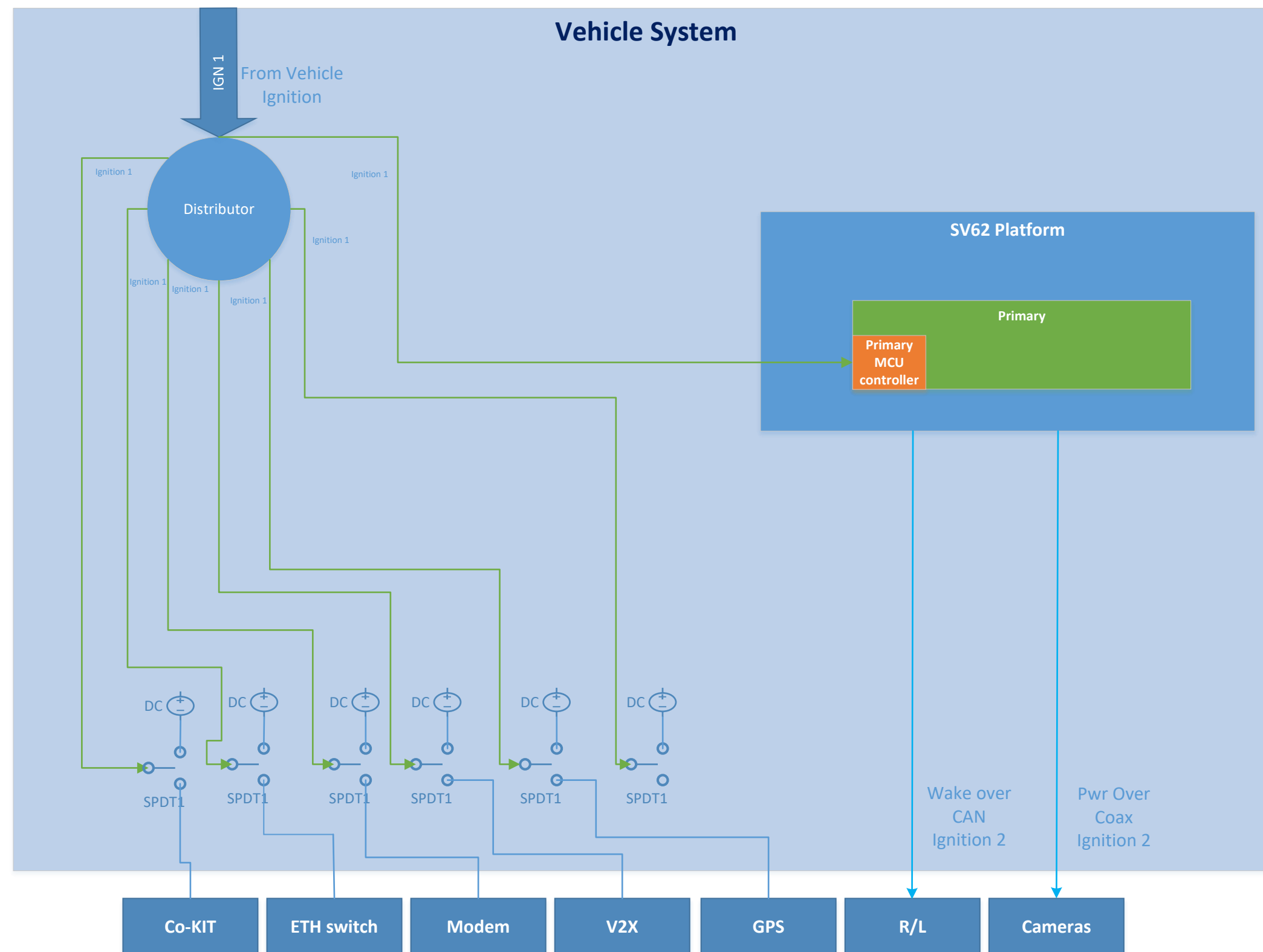
Watchdog & Reset



GPIO



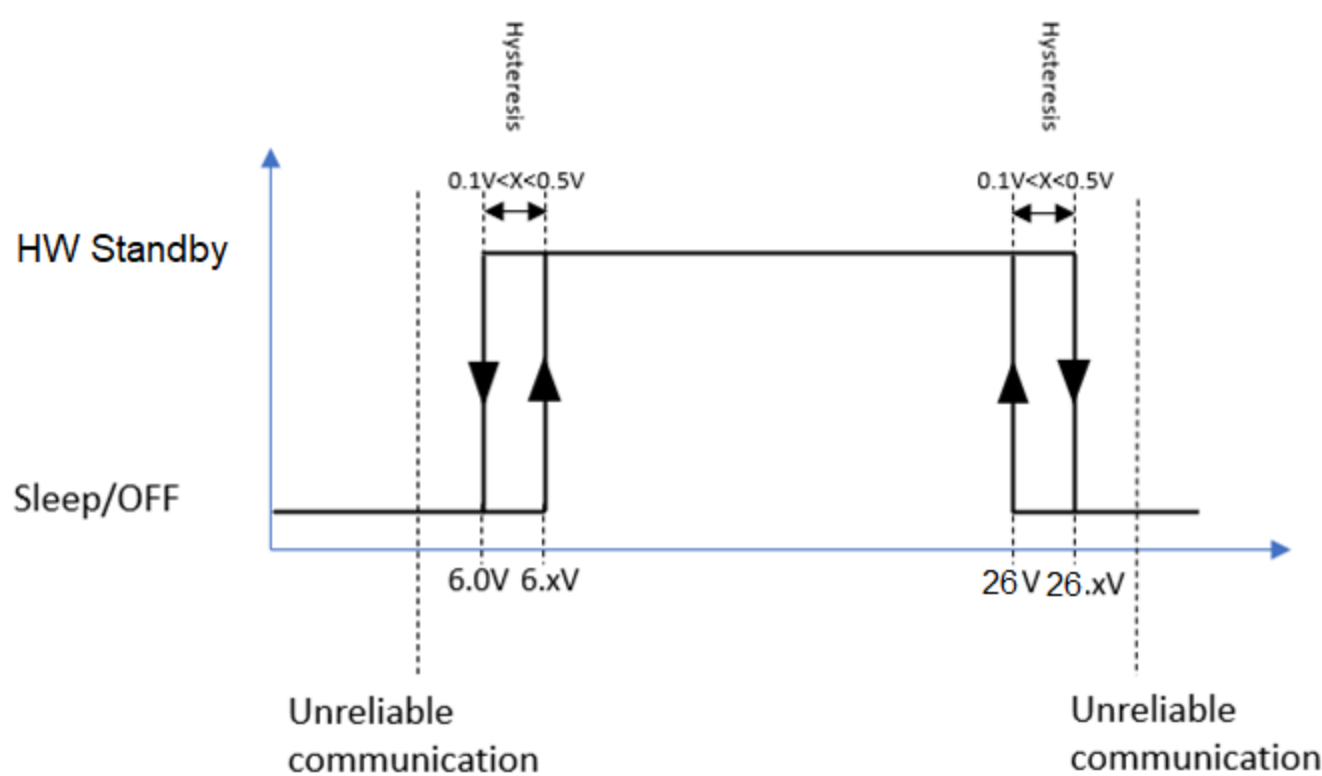
System Topology & Power-Up



Inputs Range Voltages: We must support VW8000

- 9V - 17V: Full operation (entire temperature range, without a time limit)
- 6V - 9V & 17V - 18V: All vehicle communication supported include ETH

Under/Over voltage input



HW Standby mode can be in 2 options:

1. If the inputs voltage is between 6V-9V or 17V-26V with ignition ON
2. Initiated entry to HW standby mode by software even at full operation mode (9V-17V input voltage level)

Operation voltage input

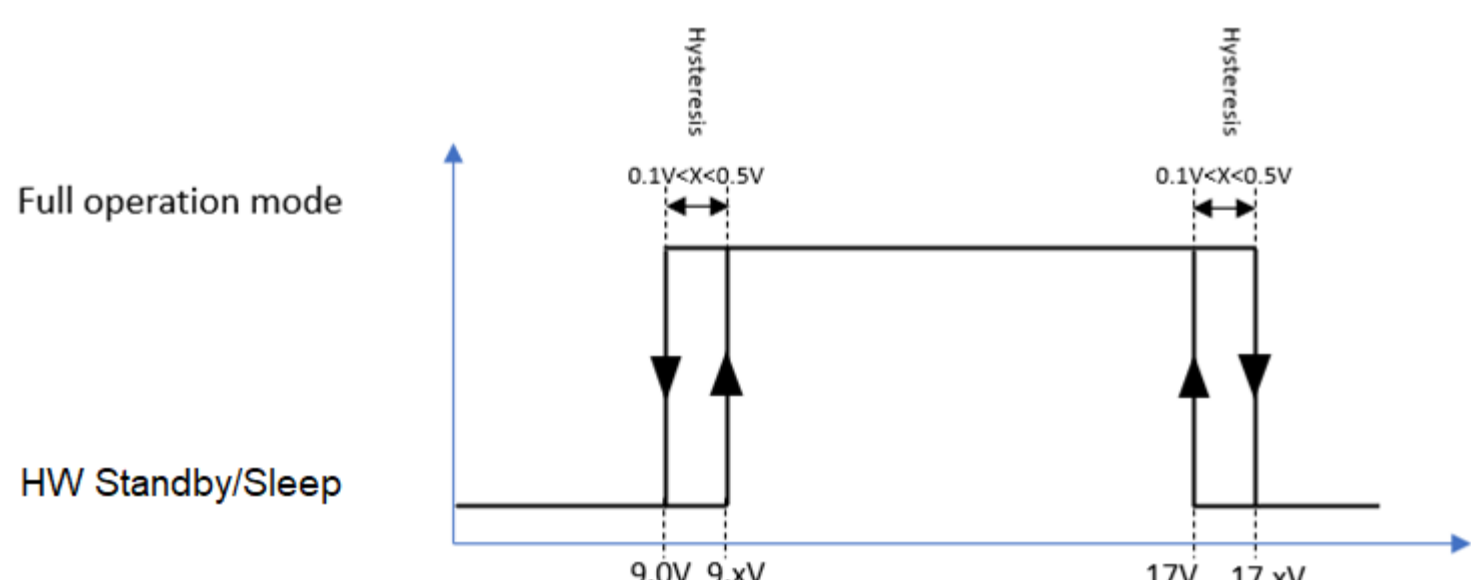


Table 1 – Function classes and operating voltage ranges									
Voltage range in V	Test	Duration	Function class						
			1	2	3	4	5	6	
17 - 26	E-04 (Jump start)	≤ 60 s	C	C	C	C	C	C	C
16 - 17	E-01 (Long-term overvoltage)	≤ 1 h	A	A	A	C	C	C	A
18 - 27	E-05 (Load dump)	≤ 300 ms	A	A	A	C	C	C	A
16 - 18	E-02 (Transient overvoltage)	See E-02	A	A	A	A	A	A	A
9,8 - 16	Static operating voltage range		A	A	A	A	A	A	⚠
9 - 9,8	Static operating voltage range		A	A	C	A	C	A	⚠
6 - 9	Static operating voltage range		A	C	C	C	C	⚠	⚠
0 - 6	Static operating voltage range		C	C	C	C	C	⚠	⚠
≥ 9	E-03a (Transient undervoltage)	500 ms	A	A	A	A	A	A	A
≥ 7	E-11 (Hot start)	See E-11	A	A	A	A	A	C	A
≥ 6	E-03b (transient undervoltage)	See E-03b	A	A	A	C	A	C	A
≥ 4,5	E-11, normal (Cold start)	See E-11	A	C	C	C	C	C	C
≥ 3,2	E-11, severe (Cold start)	See E-11	A	C	C	C	C	C	C
≥ 0	E-10 (Brief interruptions)	≤ 100 μs	A	A	A	A	A	A	A
≥ 0	E-10 (Brief interruptions)	> 100 μs	C	C	C	C	C	C	C

M: The static operating voltage range is defined in the relevant Group Bx Performance Specification

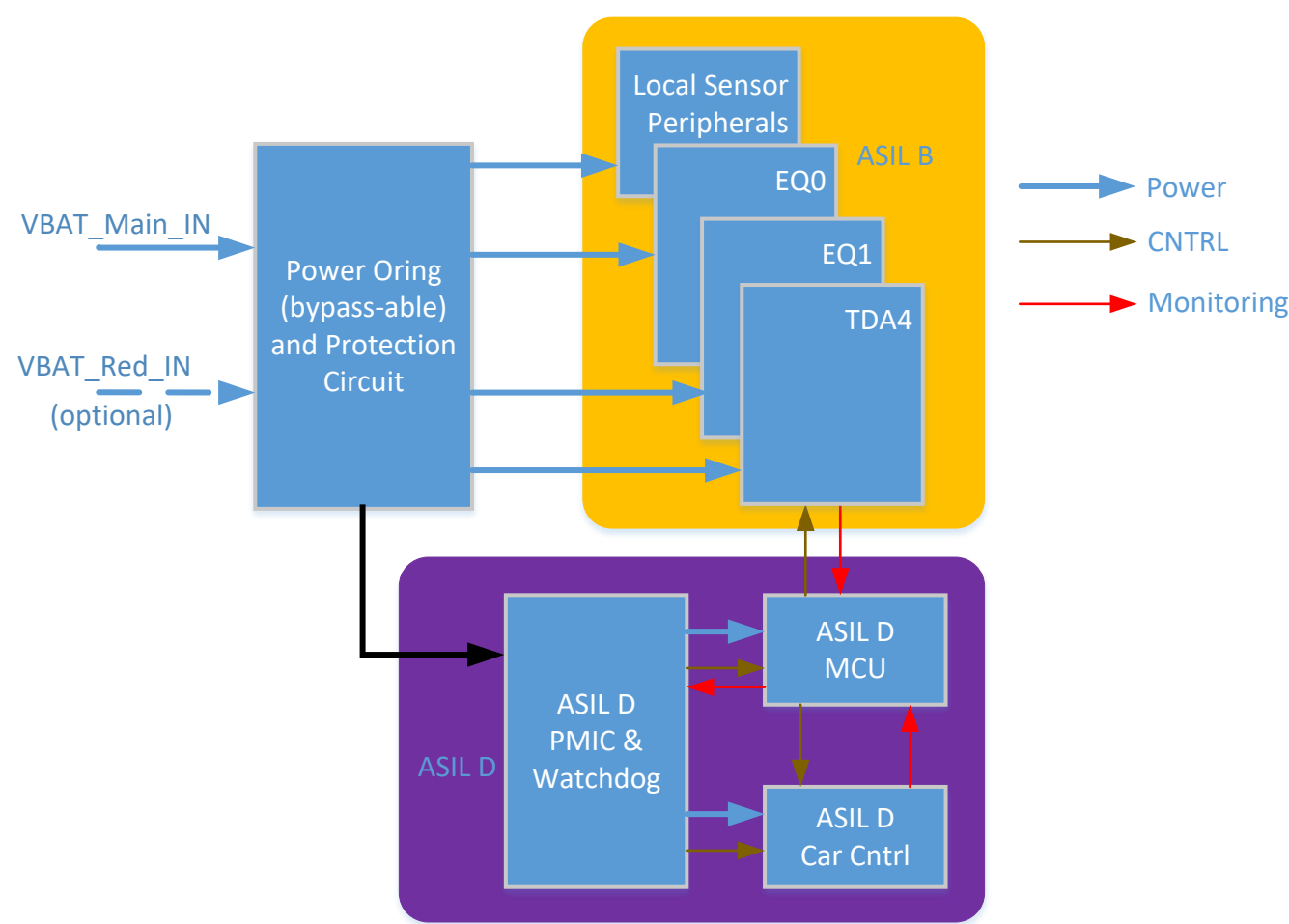
⚠ The static operating voltage range is defined in the relevant Group Bus Performance Specification.

For Vehicle CAN:

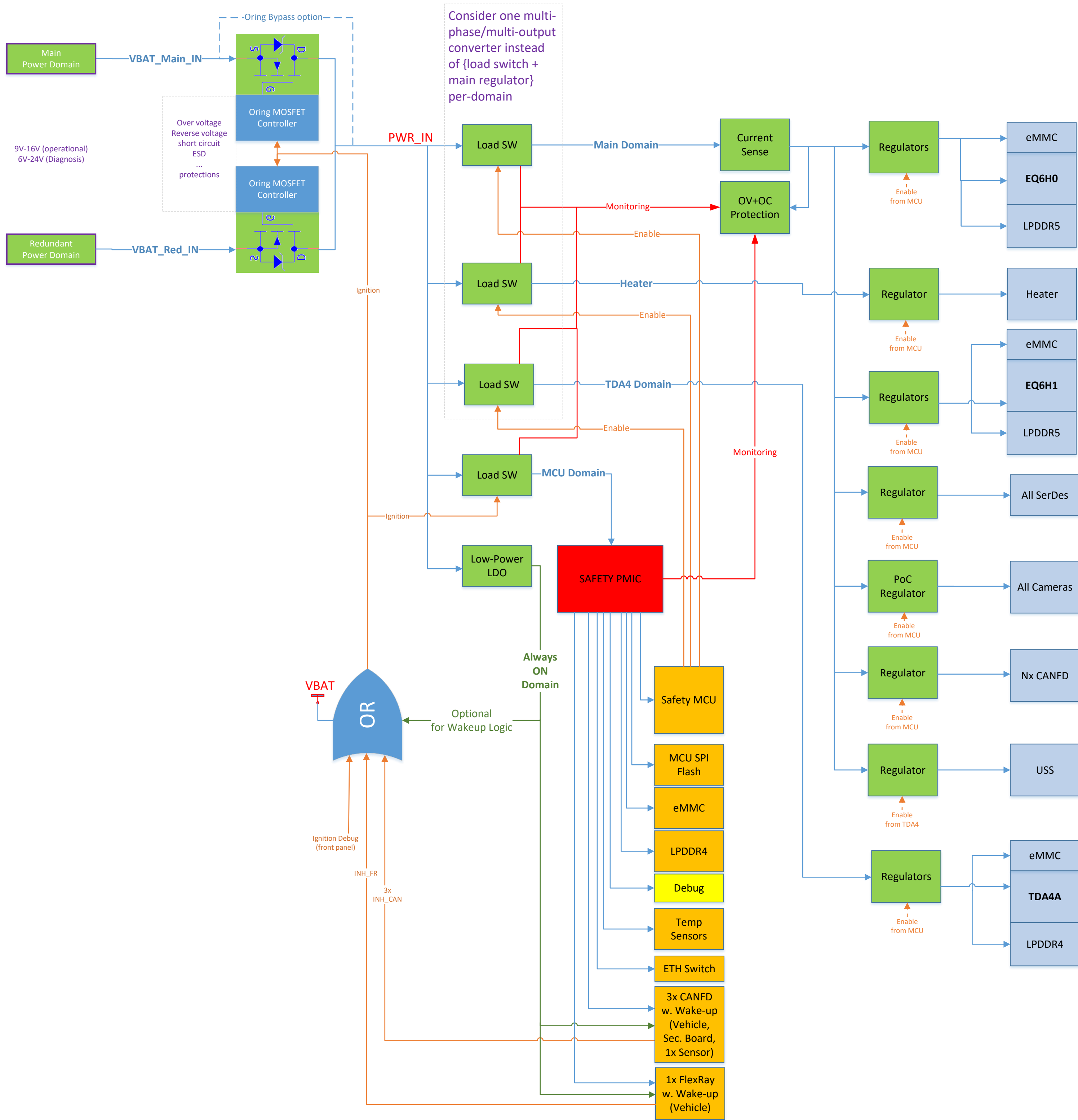
Table: Operating voltage limits

Designation	Value	Tolerance	Note
Minimum battery voltage V_{BAT_MIN}	6.5 V	±2%	Minimum static operating voltage limit
Maximum battery voltage $V_{BAT_MAX_1}$	18.0 V	±2%	Maximum static operating voltage limit

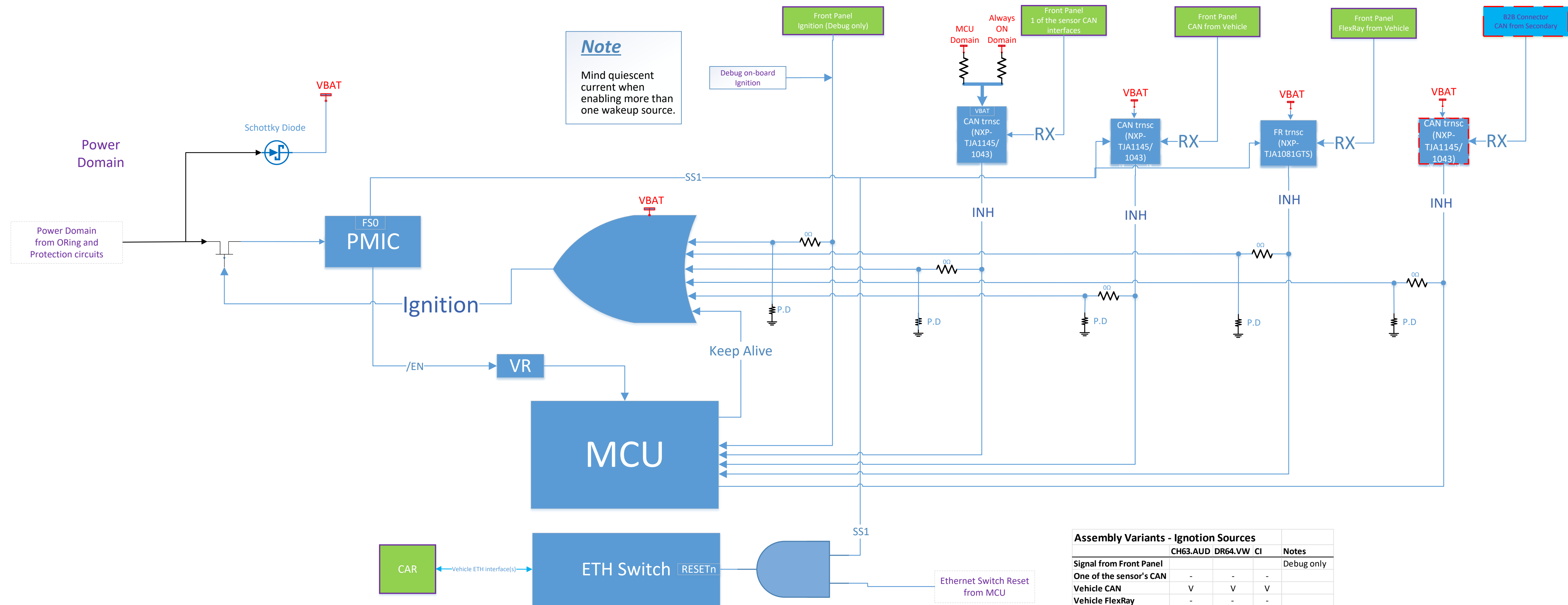
High-Level Power Topology



Power Tree suggestion



Ignition and Safe-State

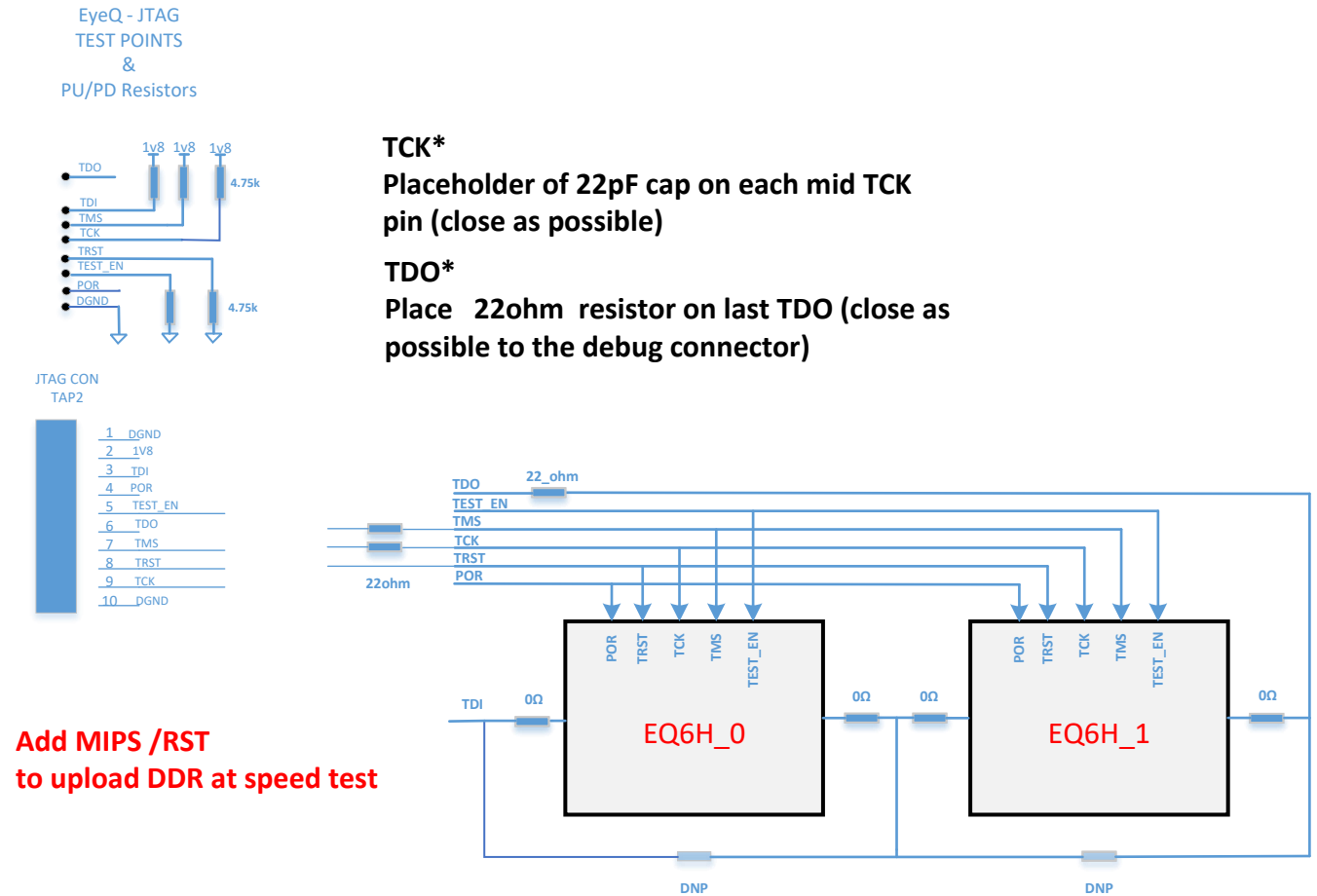


Assembly Variants - Ignition Sources				Notes
	CH63.AUD	DR64.VW	CI	
Signal from Front Panel				Debug only
One of the sensor's CAN	-	-	-	
Vehicle CAN	V	V	V	
Vehicle FlexRay	-	-	-	
B2B Connector CAN	-	-	-	

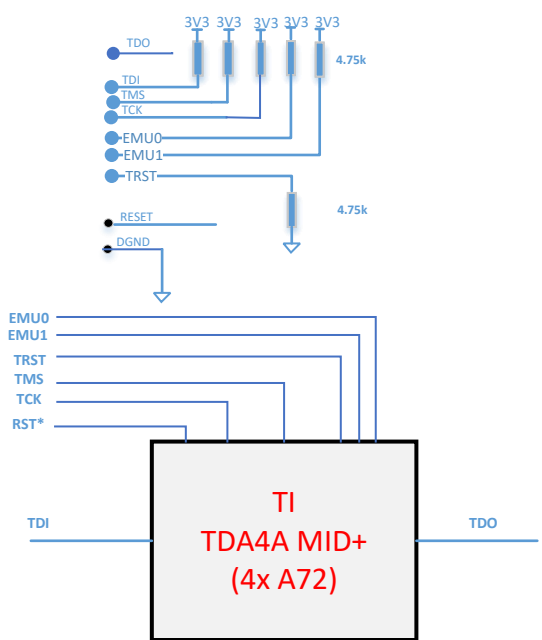
JTAG Debug & Boundary-Scan

- * Notes:**
- 1. The designer shall implement the "operational mode" as default and enable the option written below in "red" by on-board test points and on a connector in order to allow for work in JTAG mode
 - 2. The designer can choose connector pitch

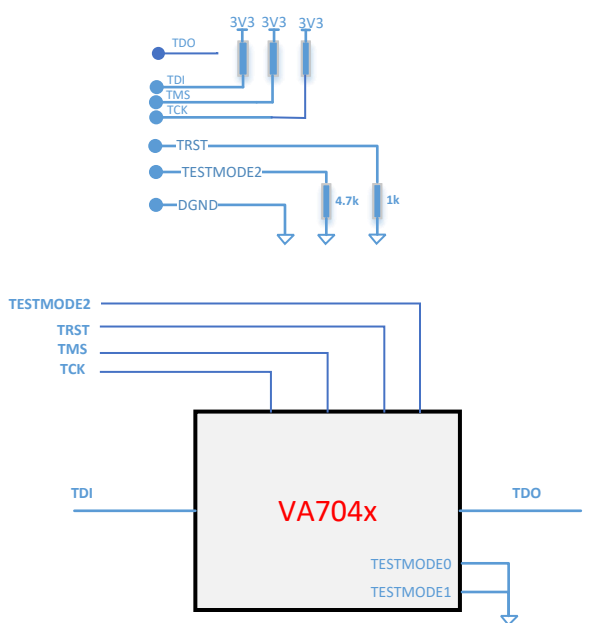
EyeQs



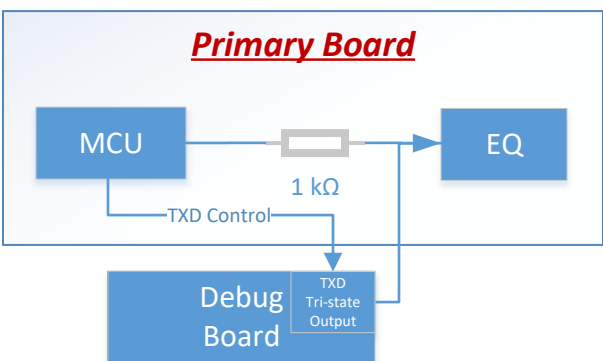
TI TDA4 MCU



Valens VA70xx Deserializers (implement per-des)

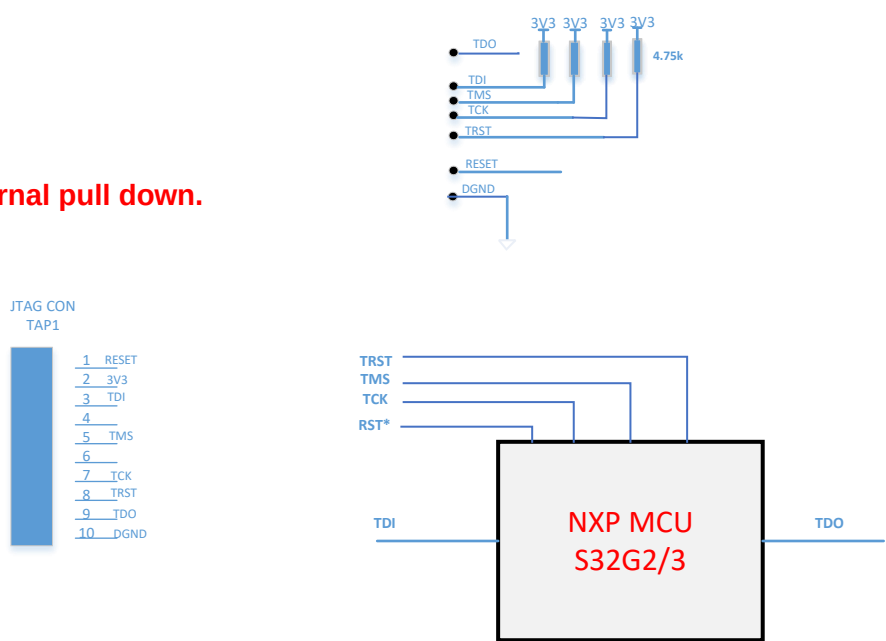


EQs UART TXD



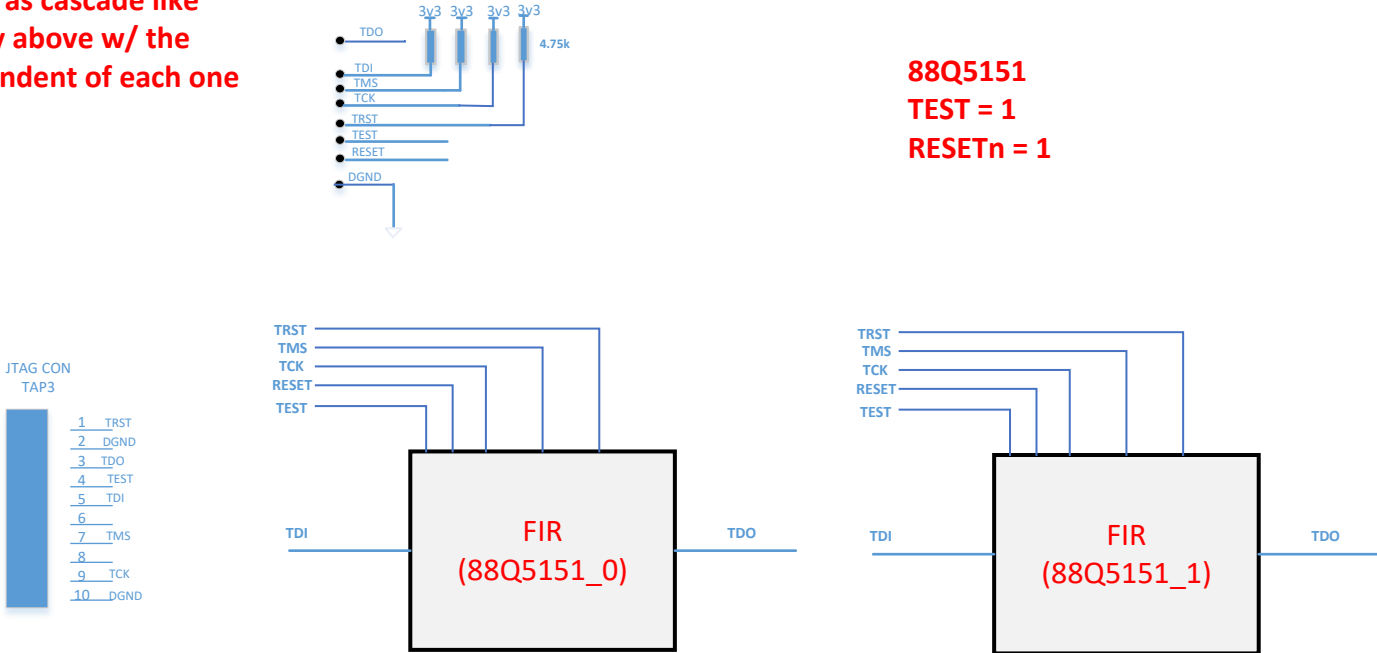
NXP MCU

- JCOMP = 1
POR_B = 1
PMIC_VDD_OK = 1
TEST = 0
TCK pin must have external pull down.



ETHSW

Implement JTAG as cascade like the EQs topology above w/ the option of independent of each one of them



- 88Q5151
TEST = 1
RESETn = 1

